

TAURANGA, New Zealand

WMO: 931860

Lat: 37.6706S

Lon: 176.1964E

Elev: 15

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.9	39.6	32.1	26.5	43.1	34.0	28.6	43.8	26.4	55.5	24.0	56.0	5.5	220	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.0	78.5	66.1	76.6	65.7	74.9	65.2	70.0	73.9	68.8	72.9	67.7	72.0	12.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	105.7	71.9	67.3	100.5	70.9	66.1	96.1	70.1	34.0	73.9	33.0	73.2	32.1	72.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	20.8	18.5	DB	34.6	83.2	1.5	2.3	33.5	84.8	32.7	86.1	31.8	87.4	30.8	89.1	(4)
(5)			WB	33.0	72.0	1.5	1.5	31.9	73.1	31.1	73.9	30.3	74.8	29.2	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	67.4	67.9	65.2	60.8	56.7	52.3	51.0	51.7	54.6	57.4	60.6	64.7	(6)	
	DBStd	6.92	3.52	3.32	3.30	3.96	4.19	4.20	3.85	3.48	3.58	3.35	3.70	3.35	(7)	
	HDD50	89	0	0	0	0	3	23	35	22	5	1	0	0	(8)	
	HDD65	2416	17	10	38	135	258	380	434	411	313	236	138	46	(9)	
	CDD50	3428	540	501	473	323	211	92	66	76	142	230	318	456	(10)	
	CDD65	280	92	92	45	8	0	0	0	0	0	0	6	37	(11)	
	CDH74	576	234	211	52	6	0	0	0	0	0	0	19	54	(12)	
	CDH80	27	13	11	1	0	0	0	0	0	0	0	1	1	(13)	
(14) Wind	WSAvg	8.6	9.1	8.4	8.1	7.5	7.7	7.9	8.1	8.1	8.9	10.1	10.2	9.4	(14)	
(15) Precipitation	PrecAvg	65.6	4.3	4.4	5.5	6.3	5.8	6.7	6.9	6.3	5.2	4.6	4.2	5.5	(15)	
	PrecMax	86.7	14.8	12.9	11.7	14.0	15.0	11.2	15.1	13.3	9.3	10.1	12.6	12.7	(16)	
	PrecMin	48.0	1.2	0.6	1.0	0.5	0.7	2.6	1.5	3.3	2.0	1.5	1.4	1.1	(17)	
	PrecStd	9.2	2.7	2.9	2.7	3.4	2.9	2.1	3.5	2.2	1.8	1.9	2.3	2.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.7	81.5	77.9	73.8	68.7	64.0	62.4	63.2	67.2	69.7	76.5	78.3	(19)	
		MCWB	67.1	67.7	65.2	64.2	62.5	59.7	57.2	55.9	58.7	59.2	63.2	64.7	(20)	
	2%	DB	78.7	78.3	75.4	71.0	66.8	62.4	60.7	61.1	64.5	67.3	72.2	75.4	(21)	
		MCWB	65.9	66.8	64.6	63.5	61.8	58.8	55.8	55.0	57.3	58.5	61.6	64.0	(22)	
	5%	DB	76.5	76.5	73.6	69.3	65.4	61.1	59.4	59.7	62.7	65.6	69.9	73.1	(23)	
		MCWB	66.0	66.3	64.3	62.7	60.8	57.4	54.9	54.4	56.5	57.8	60.8	63.5	(24)	
	10%	DB	74.6	74.8	72.1	68.0	64.1	59.8	58.1	58.4	61.1	64.0	67.8	71.3	(25)	
		MCWB	65.2	66.0	63.8	62.1	59.6	55.9	53.7	53.7	55.8	57.1	59.7	62.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.1	72.7	69.5	68.2	65.5	62.2	59.8	59.2	60.9	62.0	65.5	68.5	(27)	
		MCDB	75.2	75.9	72.4	70.4	66.7	63.1	60.9	60.6	63.7	65.3	70.6	72.4	(28)	
	2%	WB	69.6	70.8	68.3	66.3	63.9	60.2	57.8	57.4	59.4	60.7	64.0	67.3	(29)	
		MCDB	74.0	74.3	71.5	68.7	65.6	61.6	59.5	59.3	62.3	64.3	68.5	71.0	(30)	
	5%	WB	68.4	69.4	67.0	65.0	62.3	58.5	56.0	55.8	58.2	59.7	62.9	66.2	(31)	
		MCDB	73.2	73.5	70.8	67.8	64.7	60.4	58.4	58.5	61.4	63.5	67.4	70.1	(32)	
	10%	WB	67.2	68.0	65.8	63.4	60.3	56.4	54.3	54.4	57.0	58.7	61.6	65.1	(33)	
		MCDB	72.3	72.5	70.2	67.0	63.2	59.2	57.4	57.6	60.5	62.7	66.4	69.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.0	13.4	13.5	12.8	12.6	12.7	13.0	13.0	13.0	13.4	12.4	(35)	
		MCDBR	16.5	15.8	15.1	13.6	12.4	10.9	12.1	13.2	14.0	14.6	15.9	15.4	(36)	
	5% WB	MCWBR	8.1	7.6	7.8	8.0	8.1	8.0	8.6	8.8	8.9	8.8	8.5	7.9	(37)	
		MCWBR	11.3	10.9	10.5	9.5	9.1	9.0	10.0	10.4	11.2	11.5	12.2	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.349	0.348	0.328	0.313	0.309	0.302	0.297	0.302	0.329	0.336	0.335	0.352	(40)	
		taud	2.544	2.565	2.620	2.655	2.624	2.622	2.639	2.631	2.525	2.518	2.539	2.530	(41)	
	Ebn at Noon		312	305	299	285	266	259	268	285	293	305	314	312	(42)	
		Edh at Noon	34	32	28	24	21	20	21	24	30	33	34	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2092	1797	1493	1063	748	588	658	902	1220	1611	1918	1980	(44)	
	RadStd		136	154	92	75	43	46	46	51	97	108	115	163	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	-23	-173	+234	+76	(46)
(47)	Regional (4 neighbors)	+0.54	+0.67	+0.83	N/A	N/A	N/A	-18	-142	+167	N/A	(47)

Nomenclature: See separate page